

Doctor Appointment scheduler

Name: K.Kaviarasan

Course: ASP Dot Net

Project Guide: Yovan Doss X

S.NO	TOPICS	PAGE NO
1	Introduction	1.1
2	Scope	1.1
3	Objectives	1.1
4	Modules/Features	2.2
5	Workflow/ER Diagram	3.3
6	Table Design	4.5
7	Time Duration	5.6
8	Layout Design	6.1

1. Introduction

The **Doctor Appointment Booking System** is designed to enhance the healthcare appointment scheduling experience for patients, doctors, and super admins. It facilitates an efficient and user-friendly platform where patients can easily book, reschedule, and manage appointments, while doctors can view and manage their schedules. Super admins will have full control over managing doctor and patient data, appointments, and system settings. This system aims to reduce manual workload, improve efficiency, and ensure smooth management of appointments.

2. Scope

The scope of the **Doctor Appointment Booking System** is to develop an integrated platform that facilitates the management of patient appointments, doctor schedules, and administrative controls. The system will include functionalities for patient registration, appointment booking, doctor availability management, and overall appointment monitoring by the super admin.

The system will aim to:

- Provide an easy-to-use interface for patients to book, reschedule, or cancel appointments.
- Allow doctors to manage their daily appointments, apply for leave, and ensure availability.
- Enable super admins to manage doctor and patient data, oversee appointments, and implement necessary appointment blocks.

The solution will be scalable, secure, and accessible, focusing on improving appointment scheduling efficiency and user experience for all stakeholders.

3. Objectives

- **Easy Appointment Booking:**
Allow patients to easily register, book, reschedule, or cancel appointments online.
- **Manage Doctor Schedules:**
Enable doctors to view and manage their appointments, and apply for leave when needed.

- **Admin Control:**
Give super admins the ability to manage patient and doctor information, and oversee all appointments.
- **Secure Payment Integration:**
Add a secure way for patients to pay online to confirm their appointments.
- **Better Patient Experience:**
Provide patients with the ability to track their past and upcoming appointments.
- **Scalable and Secure System:**
Ensure the system is safe and can grow as more data is added.
- **Improve Efficiency:**
Automate appointment management to reduce manual work and make the system more efficient.

4. Modules / Features

1. Patient Module:

- **Registration/Login:**
 - Patients can create an account or log in using their Patient ID.
- **Appointment Booking:**
 - Select a date, department, doctor, and available time slot.
 - Payment integration for confirming appointments.
- **Appointment Management:**
 - View past, upcoming, and cancelled appointments.
 - Reschedule or cancel appointments if needed.

2. Doctor Module:

- **Login & Profile Management:**
 - Doctors can log in securely using credentials created by the Super Admin.
- **Appointment Calendar:**
 - View daily appointments in a calendar format.
 - Manage appointments by confirming, cancelling, or adjusting availability.

- **Leave Management:**
 - Doctors can apply for leave, which automatically blocks their slots from being booked during their absence.

3. Super Admin Module:

- **Master Data Management:**
 - Manage doctor and patient data (profiles, schedules, etc.).
- **Appointment Monitoring:**
 - View all scheduled appointments by doctor and patient.
- **Appointment Blocking:**
 - Block appointments for a specific doctor when needed (e.g., for emergencies or leave).
- **Doctor & Patient Management:**
 - Add, update, or delete doctor and patient profiles as needed.

4. Payment Module:

- **Payment Integration:**
 - Secure online payment gateway for patients to confirm appointments.

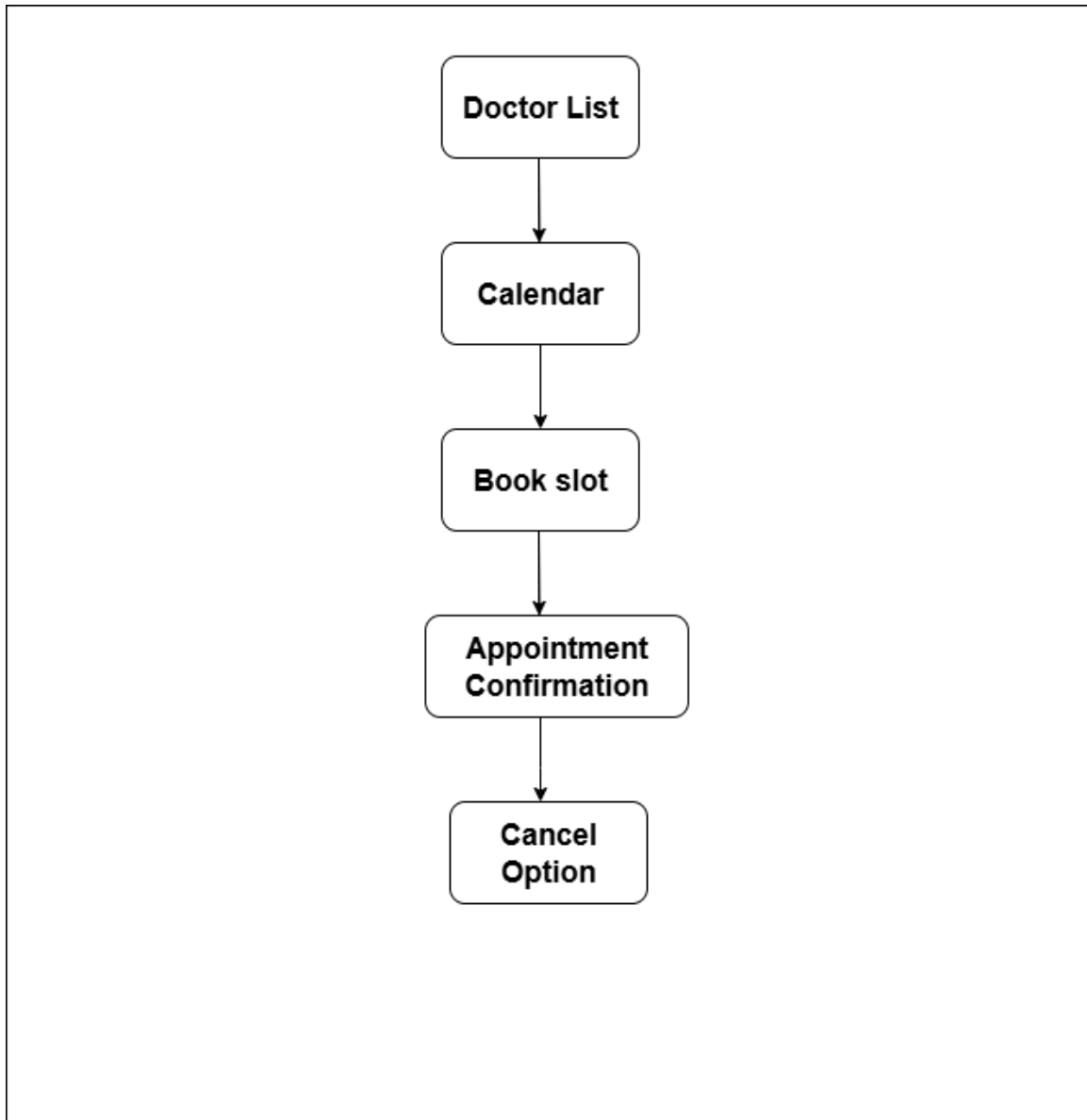
5. Notifications Module (Optional):

- **Appointment Reminders:**
 - Send automatic reminders for upcoming appointments to both patients and doctors.
- **Status Notifications:**
 - Notify patients of appointment confirmations, cancellations, or reschedules.

5. Workflow / ER Diagram

Workflow:

Doctor List → Calendar → Book Slot → Appointment Confirmation → Reschedule/Cancel



6. Table Design

Table Name: User table

Column Name	Data Type	Length	Constraint	Description

UserID	UUID	—	DEFAULT uuid_generate_v4(), PRIMARY KEY	Unique User ID (auto-generated UUID)
DoctorID	VARCHAR	50	NOT NULL	Doctor ID (string)
DoctorName	VARCHAR	255	NOT NULL	Full name of the doctor (First Name, Last Name)
Specialization	VARCHAR	100	—	Medical specialty (e.g., Cardiology, Dermatology)
Gender	ENUM	—	Values: ('Male', 'Female', 'Other')	Gender of the doctor
DateOfBirth	DATE	—	—	Doctor's date of birth
PhoneNumber	VARCHAR	15	NOT NULL	Contact number (with format validation)
Address	VARCHAR	255	—	Contact address
Username	VARCHAR	100	NOT NULL, UNIQUE	Unique username for login
Password	VARCHAR	255	NOT NULL	Secure encrypted password for login
ConfirmPassword	VARCHAR	255	NOT NULL	Re-entered password for confirmation
Role	ENUM	—	NOT NULL, Values: ('Doctor', 'Admin')	Role of the user
YearsOfExperience	INT	—	—	Number of years the doctor has been practicing

Qualification	VARCHAR	100	—	Highest qualification (e.g., MBBS, MD)
ConsultationFee	DECIMAL	(10,2)	—	Consultation fee for appointments
StartTime	TIME	—	—	Start time for doctor availability
EndTime	TIME	—	—	End time for doctor availability
isActive	BOOLEAN	—	DEFAULT TRUE	Status flag if doctor is active
isDelete	BOOLEAN	—	DEFAULT FALSE	Status flag if doctor record is deleted

Table Name:Booking Appointment table

Column Name	Type	Description
PatientId	UUID	Primary Key, auto-generated with gen_random_uuid()
TitleId	UUID	Foreign Key → Titles(TitleId) (e.g., Mr., Mrs., Dr.)
Name	VARCHAR(255)	Patient full name (required)
Gender	SMALLINT	Gender code (0=Male, 1=Female, 2=Other, etc.)
DOB	TIMESTAMP	Date of Birth
Mobile	VARCHAR(20)	Mobile number
Email	VARCHAR(255)	Email address

MaritalStatusId	UUID	Foreign Key → MaritalStatus(MaritalStatusId)
OccupationId	UUID	Foreign Key → Occupations(OccupationId)
NationalityId	UUID	Foreign Key → Nationalities(NationalityId)
State	VARCHAR(100)	State of residence
PinCode	VARCHAR(20)	Postal/ZIP code
Address	VARCHAR(500)	Full residential address
IsActive	BOOLEAN	Account status (default = TRUE)
Username	VARCHAR(100)	Unique login username
Password	VARCHAR(255)	Login password (should be hashed)

Table Name:country table

Description:

Column Name	Data Type	Length	Constraint	Description
countryId	UUID	N/A	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each record
countryName	VARCHAR	100	NOT NULL	Name of the country
isDeleted	BIT	N/A	DEFAULT FALSE	Soft delete flag (default is false)

Table Name:marital status table

Description:

Column Name	Data Type	Length	Constraint	Description
maritalStatusId	UUID	N/A	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each record
maritalStatus	VARCHAR	50	NOT NULL	Marital status (e.g., Married, Single)
isDeleted	BIT	N/A	DEFAULT FALSE	Soft delete flag (default is false)

Table Name:Relationship table

Description:

Column Name	Data Type	Length	Constraint	Description
relationshipId	UUID	N/A	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each record
relationshipName	VARCHAR	50	NOT NULL	Name of the relationship (e.g., Father, Mother)

isDeleted	BIT	N/A	DEFAULT FALSE	Soft delete flag (default is false)
-----------	-----	-----	---------------	--

Table Name:Occupation table

Description:

Column Name	Data Type	Length	Constraint	Description
occupationId	UUID	N/A	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each record
occupationName	VARCHAR	100	NOT NULL	Name of the occupation (e.g., Engineer, Doctor)
isDeleted	BIT	N/A	DEFAULT FALSE	Soft delete flag (default is false)

Table Name:Religion and Title table

Description:

Column Name	Data Type	Length	Constraint	Description
religionId	UUID	N/A	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each record
religionName	VARCHAR	100	NOT NULL	Name of the religion (e.g., Christianity, Islam)
isDeleted	BIT	N/A	DEFAULT FALSE	Soft delete flag (default is false)
Column Name	Data Type	Length	Constraint	Description
titleId	UUID	N/A	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for each record
titleName	VARCHAR	50	NOT NULL	Name of the title (e.g., Mr., Mrs., Dr.)
isDeleted	BIT	N/A	DEFAULT FALSE	Soft delete flag (default is false)

7. Time Duration (4-Day Execution Plan)

Day 1 – Database & Setup

- Install & configure SQL Server.
 - Create database schema with tables:
 - Users (patients, admins)
 - Doctors
 - Appointments
 - Define relationships & foreign keys.
 - Test DB connection with sample queries.
 - Connect SQL Server to .NET backend (Entity Framework or ADO.NET).
-

Day 2 – Backend API Setup

- Set up .NET project structure (Web API).
 - Implement authentication (basic login/signup).
 - Create API for **Doctor List** (GET doctors).
 - Use Swagger/Postman to validate endpoints.
-

Day 3 – Appointment APIs

- Build APIs for:
 - Calendar availability (GET slots).
 - Book appointment (POST).
 - Confirm appointment (PUT).
- Add validation for overlapping appointments.
- Test with sample requests in Postman.

Day 4 – Appointment Management

- Implement APIs for:
 - Reschedule appointment (PUT).
 - Cancel appointment (DELETE).
- Add status tracking (Booked, Confirmed, Rescheduled, Cancelled).
- Write unit tests for all appointment APIs.

Day 5 – Frontend (Doctor & Calendar)

- Set up frontend project (React/Angular).
- Create **Doctor List page** with search & filter.
- Create **Calendar selection UI** for appointment slots.
- Integrate with backend API for doctor data.

Day 6 – Frontend (Booking & Confirmation)

- Build booking form (patient details, reason for visit).
- Display **appointment confirmation page**.
- Show booked slot details (doctor, date, time).
- Implement form validation & error handling.

Day 7 – Integration & Testing

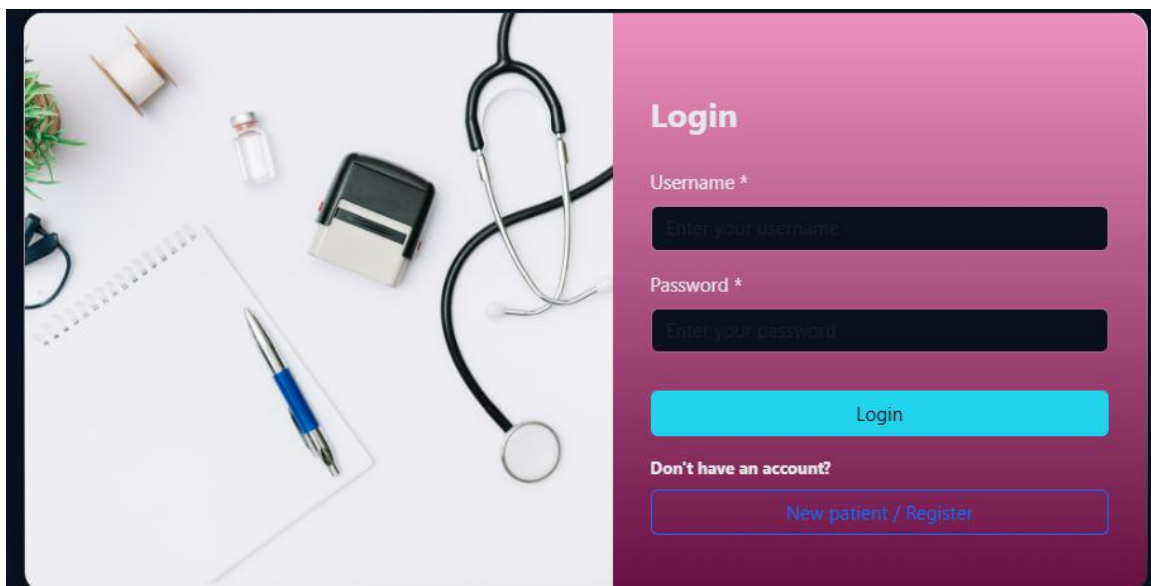
- Connect all frontend components with backend APIs.
- End-to-end flow: Doctor → Calendar → Book → Confirm → Reschedule/Cancel.
- Populate DB with demo doctors & appointments.
- Conduct functional testing with sample users.

Day 8 – Finalization & Documentation

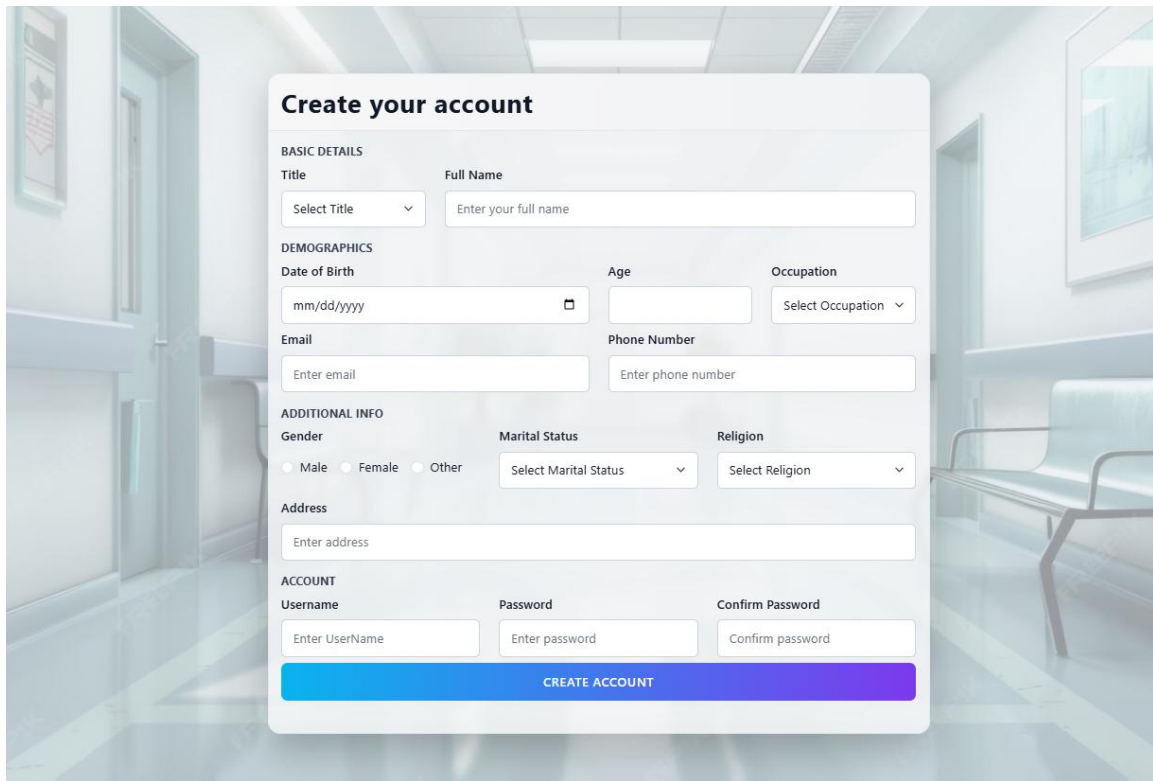
- Perform integration testing (backend + frontend).
- Fix bugs and optimize queries/APIs.
- Write project documentation:
 - ER Diagram & schema.
 - API endpoints list.
 - Screenshots of UI.
 - Setup/installation steps.
- Prepare presentation/demo version.

Layout Design:

Login Page



Register Page



Create your account

BASIC DETAILS

Title
Select Title ▾

Full Name
Enter your full name

DEMOGRAPHICS

Date of Birth
mm/dd/yyyy

Age
Enter age

Occupation
Select Occupation ▾

Email
Enter email

Phone Number
Enter phone number

ADDITIONAL INFO

Gender
☐ Male ☐ Female ☐ Other

Marital Status
Select Marital Status ▾

Religion
Select Religion ▾

Address
Enter address

ACCOUNT

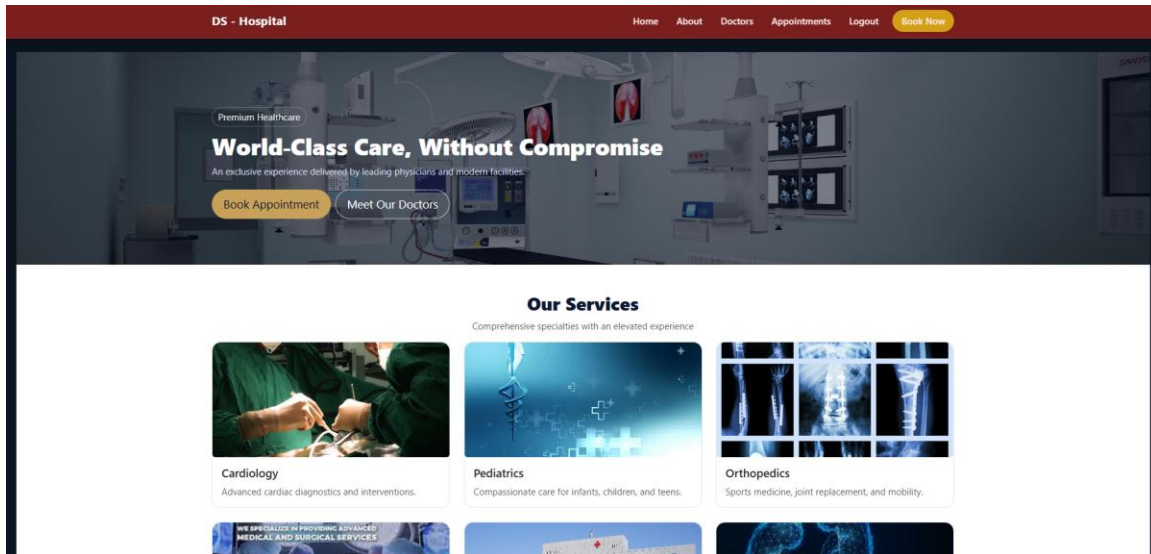
Username
Enter UserName

Password
Enter password

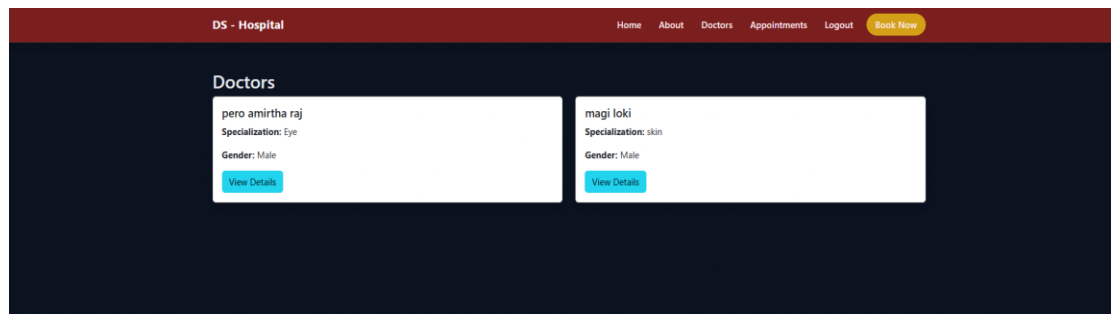
Confirm Password
Confirm password

CREATE ACCOUNT

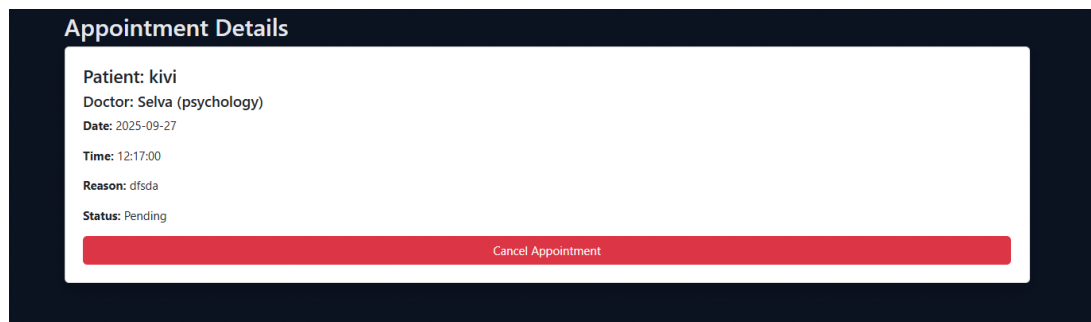
Home Page



Doctor Page



Doctor Details Page



Book Appointment Page

DS - Hospital

[Home](#)[About](#)[Doctors](#)[Appointments](#)[Logout](#)

Book Now

Book an Appointment

Simple, secure, and fast — we'll confirm your visit and send reminders.

Get Started

Patient Name

Enter full name

Age

e.g., 35

Gender

Phone

e.g., 98765 43210

Email

you@example.com

Department

Doctor

Date

Time

Message (optional)

Describe your concern

Book Appointment

Emergency Contact

24/7 Emergency Hotline

+91 44 1234 5678

Ambulance: +91 44 9876 5432

Working Hours

Mon - Fri08:00 AM - 08:00 PM

Saturday09:00 AM - 05:00 PM

SundayEmergency Only

Appointment Details

Appointment Booking Details

All Appointments						
Patient	Doctor	Specialization	Date	Time	Reason	Status
kivi	leo leo	cardiology	2025-09-25	12:00:00	fever	Pending
kivi	leo leo	cardiology	2025-09-26	13:09:00	nfsdfdfsdfsdf	Pending
kivi	leo leo	cardiology	2025-09-26	13:09:00	nfsdfdfsdfsdf	Pending
kivi	leo leo	cardiology	2025-09-26	14:45:00	viral fever	Pending
kivi	Selva Selva raj	psychology	2025-09-26	10:19:00	backbone pain	Pending
kivi	leo leo	cardiology	2025-09-26	09:23:00	osmosis	Pending
kivi	Selva Selva raj	psychology	2025-09-26	22:29:00	handbone	Pending
kivi	pero amirtha raj	Eye	2025-10-08	22:36:00	test	Pending
kivi	leo leo	cardiology	2025-09-26	10:46:00	heart pain	Pending
kivi	pero amirtha raj	Eye	2025-09-26	23:13:00	zsfdxggbhj	Pending
kivi	pero amirtha raj	Eye	2025-09-26	23:13:00	zsfdxggbhj	Pending
kivi	pero amirtha raj	Eye	2025-09-26	23:13:00	zsfdxggbhj	Pending
kivi	pero amirtha raj	Eye	2025-09-26	23:13:00	zsfdxggbhj	Pending
kivi	leo leo	cardiology	2025-09-26	12:15:00	sedrtfgyhuji	Pending
kivi	Selva Selva raj	psychology	2025-09-26	12:19:00	new	Pending
kivi	pero amirtha raj	Eye	2025-09-27	10:55:00	efsd	Pending

Appointment Cancel Page

Appointment Details

Patient: kivi
Doctor: Selva (psychology)
Date: 2025-09-27
Time: 12:17:00
Reason: dfsda
Status: Pending

Cancel Appointment